

Stock Price Prediction Using LightGBM Method with Technical Indicators

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Abstract—Stock investment offers high profit potential but comes with significant risks due to market volatility. The dynamic movement of prices makes stock price prediction both a challenge and a strategic opportunity for investors. This study aims to predict the stock price of NVIDIA Corporation (NVDA) using the Light Gradient Boosting Machine (LightGBM) method. LightGBM is selected for its strong performance on time-series data and computational efficiency. Historical stock data are obtained from Yahoo Finance covering the period from 2020 to 2025. To enhance predictive performance, several technical indicators are integrated as additional features, including the Relative Strength Index (RSI), Rate of Change (ROC), Stochastic Oscillator, Moving Average Convergence Divergence (MACD), and Bollinger Bands. The model is trained using a delta-based target approach, which predicts daily price differences and reconstructs the closing price for evaluation with Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2). The experimental results demonstrate that the hyperparameter tuned LightGBM model effectively captures price patterns, achieving a test RMSE of 3.5196 and R^2 of 0.9823, the best scenario among the three scenarios. This performance confirms LightGBM as an accurate and efficient framework suitable for real-world stock price forecasting tasks.

Keywords—stock price prediction, LightGBM, technical indicators, NVIDIA

I. INTRODUCTION

The stock market serves as a vital component of the global financial system, facilitating capital flow and investment decisions that reflect economic stability and investor sentiment. According to Frank K. Reilly [1], the ability to analyze and forecast stock price movements is fundamental to supporting rational portfolio management and minimizing investment risk. In modern markets, stock price behavior is influenced by complex economic and behavioral factors such as inflation, investor confidence, and market sentiment. Jae H. Kim and Heajin H. Ryoo [2] found that common stocks can serve as an inflation hedge over the long term. This finding emphasizes the need for accurate forecasting models capable of adapting to macroeconomic fluctuations.

Traditional time-series models such as ARIMA and ARIMA-GARCH, as well as deep learning architectures including LSTM and MLP, have been widely used for stock prediction, including studies focusing on NVIDIA Corporation (NVDA) [3], [4]. Although these models can be effective, they often require large datasets and substantial computational resources, making them less practical for lightweight or real-time forecasting systems. Ensemble learning methods, such as Gradient Boosting Decision Trees (GBDT), offer an alternative approach. Light Gradient Boosting Machine (LightGBM), introduced by Ke et al. [5], is an efficient variant of GBDT that provides strong performance on tabular and time-series data with lower computational cost.

Prior work has also demonstrated its effectiveness for stock prediction tasks [6], [7].

A key limitation in many forecasting studies is the direct prediction of absolute closing prices, which tend to follow a near-deterministic pattern where $P_t \approx P_{t-1}$. This can result in high apparent accuracy that primarily reflects trend persistence rather than genuine predictive capability. To address this issue, the present study predicts daily price changes or delta close (ΔClose) instead of absolute values, following insights from return-based financial modeling [8], [9]. This transformation reduces non-stationarity, emphasizes short-term movements, and provides a more realistic evaluation of predictive performance.

Stock price dynamics are also influenced by technical indicators that capture market momentum, volatility, and trend strength. Indicators such as Relative Strength Index (RSI), Rate of Change (ROC), Stochastic Oscillator, Moving Average Convergence Divergence (MACD), and Bollinger Bands have been widely used to enhance financial prediction models [10], [11], [12], [13], [14]. Combining these indicators with lag-based features allows the model to incorporate diverse market signals during learning.

In addition to traditional ensemble models, recent studies have explored hybrid or deep sequence-based architectures, such as CatBoost-LSTM hybrids [15] and Transformer-based predictors [16]. While these models achieve strong performance, they require complex architectures and high computational cost. This study aims to provide a simpler and more interpretable alternative by combining LightGBM, Optuna-based hyperparameter tuning, and a delta-based approach using both lag features and technical indicators.

This study contributes in three ways. First, it adopts a delta-based target formulation to focus on daily price movement rather than trend persistence. Second, it integrates LightGBM with Optuna's Tree-structured Parzen Estimator (TPE) optimization to achieve stable and reproducible performance. Third, it evaluates the influence of five technical indicators through a structured scenario-based experiment, supported by feature-importance and SHAP-based interpretability analysis.

The remainder of this paper is structured as follows. Section II presents theoretical background and related work. Section III details the methodology, covering data collection, feature engineering, model design, experimental scenarios, and the discussion of results and limitations. Section IV provides conclusion and suggestions for future works.

II. METHOD

A. Stock Market and Time-Series Data

Stock prices are organized as time-series data in which each observation is sequentially dependent on previous values. This temporal relationship often results in strong

autocorrelation, making traditional linear models less effective when capturing nonlinear market behavior [4]. To address this, machine learning methods are increasingly used to model complex stock price movements [7], [9], [10], [15], [16].

In this study, the dataset consists of daily closing prices of NVDA retrieved from Yahoo Finance [17]. Missing values were removed, and numerical fields were standardized to improve training stability.

Instead of predicting the absolute price, the model predicts the ΔClose , defined as

$$\Delta\text{Close}_t = \text{Close}_t - \text{Close}_{t-1}. \quad (1)$$

This delta-based formulation reduces trend persistence, improves stationarity, and forces the model to learn the actual market movement rather than simply approximating P_t with P_{t-1} [8], [9]. The final predicted price is reconstructed as

$$\widehat{\text{Close}}_t = \text{Close}_{t-1} + \widehat{\Delta\text{Close}}_t. \quad (2)$$

All evaluation metrics are computed using the reconstructed closing prices.

B. Gradient Boosting Decision Trees and LightGBM

Gradient Boosting Decision Trees (GBDT) are ensemble models that combine weak learners—typically decision trees—by sequentially minimizing the residual errors of previous trees [18], [19]. GBDT has shown strong performance in financial prediction tasks due to its robustness and flexibility.

LightGBM is a high-performance implementation of GBDT that introduces several optimizations to improve both speed and accuracy [5]. Its key features include:

- **Histogram-based learning**, which reduces computation time by discretizing continuous features.
- **Leaf-wise tree growth**, enabling more accurate tree structures on tabular data compared to level-wise growth.
- **Gradient-based One-Side Sampling (GOSS)**, which retains instances with larger gradients (higher loss) and subsamples those with smaller gradients.
- **Exclusive Feature Bundling (EFB)**, which merges mutually exclusive sparse features to reduce dimensionality.

These characteristics make LightGBM suitable for time-series forecasting tasks, including stock price prediction [6], [7].

To further improve model performance, Optuna was used for hyperparameter tuning, optimizing parameters such as `num_leaves`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, and regularization terms [20]. The optimization objective was to minimize the validation loss metric. All models were trained using a fixed random seed of 42, with all relevant LightGBM seed parameters (e.g., `random_state`, `seed`, `bagging_seed`) set consistently to ensure reproducibility.

C. Technical Indicators

Technical indicators provide additional information related to momentum, volatility, and market trends. These

indicators have been widely used to enhance stock prediction models [10]. Five indicators were incorporated in this study: RSI, ROC, Stochastic Oscillator, MACD, and Bollinger Bands, each computed directly from the closing-price series.

1) Relative Strength Index (RSI)

The RSI measures recent price gains relative to losses to identify overbought or oversold conditions [11]. Using a 14-day period, it is defined as

$$\text{RSI} = 100 - \frac{100}{1 + \text{RS}}, \text{RS} = \frac{\text{Average Gain}}{\text{Average Loss}}. \quad (3)$$

Values above 70 indicate overbought conditions, whereas values below 30 indicate oversold momentum.

2) Rate of Change (ROC)

The ROC represents the percentage change between the current price and the price n periods earlier [12]:

$$\text{ROC}_t = \frac{P_t - P_{t-n}}{P_{t-n}} \times 100, \quad (4)$$

where P_t and P_{t-n} denote the current and n -period past closing prices.

3) Stochastic Oscillator

The Stochastic Oscillator evaluates the position of the current price relative to the recent high–low range [13]:

$$\%K_t = \frac{P_t - L_n}{H_n - L_n} \times 100, \quad (5)$$

$$\%D_t = \text{SMA}_3(\%K_t), \quad (6)$$

where H_n and L_n are the highest and lowest prices over the past n days, and SMA_3 is a 3-period simple moving average.

4) Moving Average Convergence Divergence (MACD)

MACD captures momentum based on the difference between fast and slow exponential moving averages (EMAs) [14]:

$$\text{MACD}_t = \text{EMA}_{\text{fast}}(P_t) - \text{EMA}_{\text{slow}}(P_t), \quad (7)$$

$$\text{Signal}_t = \text{EMA}_{\text{signal}}(\text{MACD}_t), \quad (8)$$

$$\text{Histogram}_t = \text{MACD}_t - \text{Signal}_t. \quad (9)$$

A positive MACD indicates upward momentum, and crossovers between the MACD and signal line serve as potential trading signals.

5) Bollinger Bands (BB)

Bollinger Bands measure price volatility using a 20-day moving average and standard deviation σ [11]:

$$\text{Upper}_t = \text{SMA}_{20}(P_t) + 2\sigma, \quad (10)$$

$$\text{Lower}_t = \text{SMA}_{20}(P_t) - 2\sigma, \quad (11)$$

$$\%B_t = \frac{P_t - \text{Lower}_t}{\text{Upper}_t - \text{Lower}_t}. \quad (12)$$

The Bollinger Bands help detect breakout patterns and volatility extremes. All indicators were included alongside the lag features.

D. Research Flowchart and Experimental Scenarios

The research process consists of sequential stages integrating data acquisition, feature construction, model training, and evaluation.

1) *Data Collection*: Retrieve daily NVDA closing prices (2020-2025) from Yahoo Finance.

2) *Feature Engineering*: Compute lag features (1, 3, and 7 days) and the five selected indicators (RSI, ROC, Stochastic, MACD, Bollinger Bands).

3) *Data Splitting*: Apply chronological split—64% training, 16% validation, 20% testing.

4) *Model Training*:

- Scenario 1 (Basic): LightGBM using only lag features.
- Scenario 2 (Tuned): LightGBM with Optuna hyperparameter search applied to lag features.
- Scenario 3 (Combo): LightGBM using lag features plus all technical indicators, with tuned hyperparameters.

5) *Prediction and Price Reconstruction*: Predict ΔClose and reconstruct closing prices using (2).

6) *Evaluation and Interpretation*: Compute evaluation metrics; analyze feature importance and SHAP values.

E. Evaluation Metrics

To assess prediction accuracy, three metrics were employed—Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2). These metrics are computed by comparing the actual daily closing prices against the reconstructed closing prices derived from the predicted ΔClose .

1) *Root Mean Squared Error (RMSE)*: Evaluates the model's overall fit by measuring the square root of the average squared differences between predicted and actual values. A lower RMSE indicates higher accuracy and stronger robustness to large errors [21].

2) *Mean Absolute Error (MAE)*: Quantifies the average magnitude of prediction errors, showing how far the predictions deviate from the actual prices on average. Unlike RMSE, MAE is less sensitive to extreme values [22].

3) *Coefficient of Determination (R^2)*: R^2 measures the proportion of variance in the actual prices that is explained by the model. An R^2 value close to 1 signifies that the model effectively captures the underlying market patterns and delivers strong predictive reliability [23].

III. EXPERIMENT AND ANALYSIS

A. Dataset Overview

Following the methodology described in Section II, this study uses NVDA's daily closing price data from January 1, 2020, to January 1, 2025, as illustrated in Fig. 1, to train and evaluate the model.

Target variable is thus formulated as ΔClose to emphasize short-term movement signals and improve stationarity, as

shown in (1). The final predicted price is then reconstructed using the previous day's closing price and the predicted change as detailed in (2). The dataset is chronologically partitioned into non-overlapping sets: 64% for training, 16% for validation, and 20% for testing.

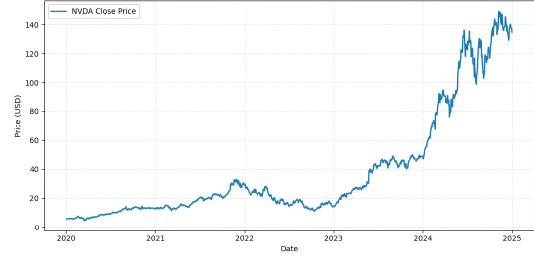


Fig. 1. Historical closing price of NVDA.

B. Experimental Setup

Experiments were executed on a system featuring an AMD Ryzen 5 5600H CPU and 16 GB of RAM, running the Windows 11 operating system. The software environment utilized Python 3.13, with key libraries including LightGBM 4.6.0 for modeling and Optuna 4.4.0 for hyperparameter optimization. To ensure reproducibility, the LightGBM model was configured with all relevant seed parameters (e.g., `random_state`, `seed`, `bagging_seed`) set to 42, alongside deterministic settings. Three distinct feature scenarios were tested:

- **Basic**: lagged price features (lag_1, lag_3, and lag_7) without hyperparameter tuning.
- **Tuned**: the same lag features with Optuna-based hyperparameter optimization.
- **Combined Indicators (Combo)**: lag features augmented with the five standard technical indicators.

The best hyperparameters identified during the Tuned scenario were extracted from the Optuna search and applied to the Combo scenario. The key values include:

- **Tree structure**: `num_leaves` = 115, `max_depth` = 9, `min_child_samples` = 88.
- **Learning**: `learning_rate` = 0.0797.
- **Regularization**: `reg_alpha` = 0.3120, `reg_lambda` = 0.2904, and `min_split_gain` = 0.6011.
- **Subsampling**: `subsample` = 0.8395, `colsample_bytree` = 0.6624, `feature_fraction` = 0.6082, with `bagging_freq` = 7.

C. Model Performance

Table I reports the testing performance for the three scenarios.

TABLE I. MODEL PERFORMANCE ON TEST SET

Scenario	RMSE	MAE	R^2
Basic	3.5829	2.7285	0.9817
Tuned	3.5196	2.6475	0.9823
Combo	3.7165	2.9031	0.9796

Tuned scenario achieved the best overall performance across all evaluation metrics, indicating that Optuna's

hyperparameter search improved the model's generalization. The Combo scenario produced competitive results but showed mild degradation compared to the Basic and Tuned models, which may be attributed to indicator redundancy and increased feature noise. Further insight into the forecasting behavior is provided in Fig. 2, which plots the actual prices series against the predictions across all scenarios.

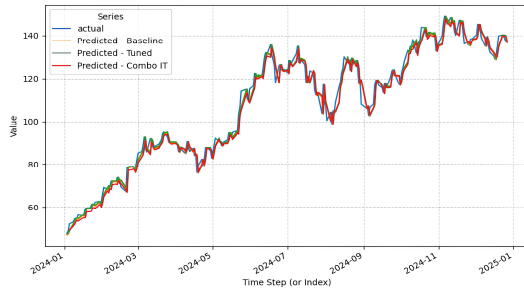


Fig. 2. Actual versus predicted closing prices for all scenarios.

As shown in Fig. 2, the Tuned model aligns most closely with the actual price series, particularly during volatile periods. The Basic model lags during rapid fluctuations, while the Combo model follows the overall trend but exhibits mild overfitting around mid-2024.

D. Feature Importance

To understand the contribution of each input variable, two interpretability tools were used: LightGBM feature importance and SHAP analysis. Fig. 3 presents the ranking of feature importance as calculated by the LightGBM algorithm, and Fig. 4 shows the global and local feature impacts derived from the SHAP beeswarm plot.

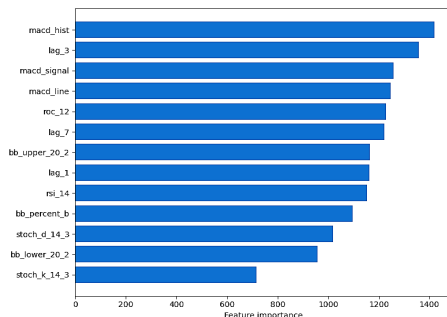


Fig. 3. Feature importance ranking.

MACD-related features—particularly `macd_hist`—and lag features (`lag_3` and `lag_7`) appear among the most influential predictors. This suggests that short-term price memory and momentum-based indicators play a dominant role in the model's decision process.

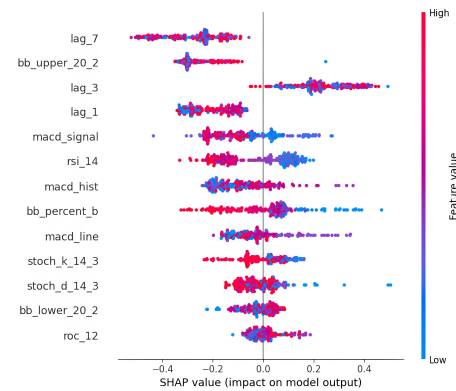


Fig. 4. SHAP beeswarm plot showing global and local feature impact.

The SHAP beeswarm plot confirms the prominence of lag features and the Bollinger upper band (`bb_upper_20_2`). High values of `lag_7` consistently push predictions upward, whereas low values of `bb_upper_20_2` exert a downward effect on the output. MACD signal (`macd_signal`) and RSI also exhibit clear directional influence. The strong agreement between LightGBM feature importance and SHAP results further strengthens confidence in the model's interpretability [24].

E. Discussion of Results

Three key observations emerged from the experimental analysis. First, the efficacy of Optuna tuning was confirmed, as the resulting model consistently achieved the lowest error metrics, demonstrating the effectiveness of automated hyperparameter optimization. Second, the integration of technical indicators introduced both information and noise. The Combined feature set did not outperform the optimized lag-only feature set, primarily because some indicators, notably the Bollinger lower band and ROC, exhibited lower SHAP weights. This behavior aligns with findings from previous studies on feature-set complexity in financial forecasting [10], [11], [13].

Third, the delta-based formulation proved critical for realistic short-term forecasting. By predicting the ΔClose rather than absolute prices, the model avoids the inflated accuracy typically caused by trend persistence and is instead forced to capture actual short-term market fluctuations [8], [9]. The superiority of this approach is particularly visible in Fig. 2 during the sharp market movements observed in mid-2024.

Compared with more complex ensemble and deep sequence-based methods (e.g., CatBoost-LSTM hybrids or Transformer-based models) [15], [16], the proposed LightGBM-delta framework achieves competitive performance while remaining computationally efficient and easier to deploy and interpret in practice.

F. Limitations and Considerations

Although the model performs well, several limitations remain. LightGBM does not explicitly model sequential dependencies, making it less sensitive to long-range temporal patterns compared to LSTM or Transformer models. Technical indicators can introduce noise during highly volatile periods, and real-world deployment would require continuous or periodic retraining to address market regime shifts.

Moreover, this study evaluates only one-step-ahead forecasts; multi-step forecasting remains for future work.

Practical implementation in live trading or decision-support systems would also need to account for transaction costs, execution latency, and potential data delays, which are not modeled in this study.

IV. CONCLUSION

This study examined the application of LightGBM for short-term stock movement prediction using NVDA's daily closing prices. Three scenarios were evaluated: a Basic scenario using only lag features, a Tuned scenario incorporating Optuna-based hyperparameter optimization, and a combo scenario that augmented lag features with multiple technical indicators. The results showed that the Tuned model achieved the best performance across RMSE, MAE, and R^2 , demonstrating the effectiveness of automated hyperparameter search. Although the combo scenario introduced additional information, it did not outperform the Tuned model due to feature redundancy and indicator-related noise.

Interpretability analysis further supported these findings. Feature importance rankings identified lag features and MACD components as the top influential predictors, highlighting the importance of short-term price memory and momentum signals. The SHAP beeswarm plot reinforced the dominance of lag features—particularly lag₇—and revealed consistent directional effects for the Bollinger upper band, MACD signal, and RSI. The strong alignment between LightGBM feature importance and SHAP values increases confidence in the model's interpretability and reliability.

Delta-based formulation also proved beneficial. By predicting the one-day change in price rather than the absolute closing value, the model avoided the inflated accuracy associated with strong price autocorrelation and was able to capture short-term fluctuations more effectively, especially during the sharp market movements observed in mid-2024.

Despite these strengths, several limitations remain. LightGBM does not capture long-range temporal dependencies, and some technical indicators introduce noise during highly volatile periods. Real-world deployment would require periodic retraining to adapt to evolving market regimes and explicit handling of transaction costs and latency. Moreover, this study focuses solely on one-step-ahead forecasting. Future work may explore multi-step forecasting, the incorporation of macroeconomic or sentiment-based features, or hybrid architectures that combine tree-based models with sequence-aware methods.

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